



testbench.sv

```

1 module TB;
2
3 reg [3:0] A,B;
4 reg [2:0] OP;
5 wire [3:0] alu_out;
6
7 alu a1(A,B,OP,alu_out);
8
9 initial
10 begin
11 $monitor("OP=%b A=%b B=%b OUT=%b", OP, A, B, alu_out);
12
13 OP = 3'b000; A = 4'b0011; B = 4'b0001;
14 #10;
15
16 OP = 3'b001; A = 4'b0011; B = 4'b0001;
17 #10;
18
19 OP = 3'b010; A = 4'b0011; B = 4'b0001;
20 #10;
21
22 OP = 3'b011; A = 4'b0011; B = 4'b0001;
23 #10;
24

```

design.sv

```

4 input [2:0] OP;
5 output reg [3:0] alu_out;
6
7 always @(*)
8 begin
9     case(OP)
10
11         3'b000: alu_out = 0;
12         3'b001: alu_out = A + B;
13         3'b010: alu_out = A - B;
14         3'b011: alu_out = A & B;
15         3'b100: alu_out = A | B;
16         3'b101: alu_out = ~A;
17         3'b110: alu_out = ~B;
18         3'b111: alu_out = 0;
19
20     default: alu_out = 0;
21
22     endcase
23 end
24
25 endmodule

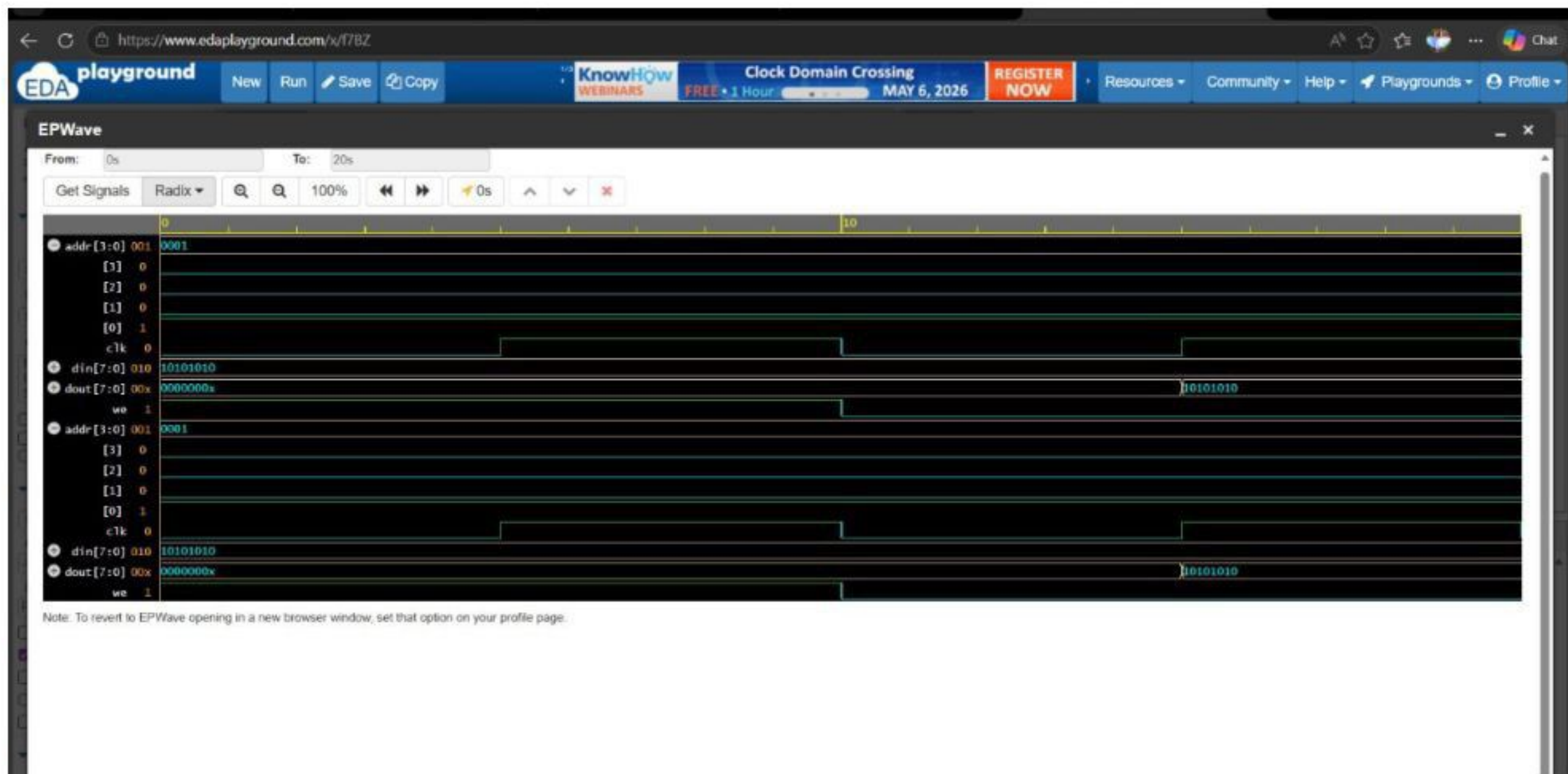
```

Log

Share

[2026-05-14 07:18:46 UTC] iverilog -wall -g2012 design.sv testbench.sv && unbuffer vvp a.out
OP=000 A=0011 B=0001 OUT=0000
OP=001 A=0011 B=0001 OUT=0100
OP=010 A=0011 B=0001 OUT=0010
OP=011 A=0011 B=0001 OUT=0001
testbench.sv:29: \$finish called at 50 (1s)
Done

Task 1(arithmetic logic unit (ALU))



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Options

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n.bash shell script

EPWave after run

output text file

output SVG file

oad files after run

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EDA Playground

testbench.sv

design.sv

```
1 reg clk;
2 reg we;
3 reg [3:0] addr;
4 reg [7:0] din;
5 wire [7:0] dout;
6
7 ram uut (
8     .clk(clk),
9     .we(we),
10    .addr(addr),
11    .din(din),
12    .dout(dout)
13 );
14
15 always #5 clk = ~clk;
16 initial begin
17     $dumpfile("dump.vcd");
18     $dumpvars(0, ram_tb);
19 end
20 initial
21 begin
22     clk = 0;
23 end
```

```
1 module ram(
2     input clk,
3     input we,
4     input [3:0] addr,
5     input [7:0] din,
6     output reg [7:0] dout
7 );
8
9 reg [7:0] memory [15:0];
10
11 always @(posedge clk)
12 begin
13     if (we)
14         memory[addr] <= din;
15
16     dout <= memory[addr];
17 end
18
19 endmodule
```

Log

Share

```
[2026-05-15 05:05:23 UTC] verilog '-wall' '-g2012' design.sv testbench.sv && unbuffer vvp a.out
VCD info: dumpfile dump.vcd opened for output.
testbench.sv:38: $finish called at 20 (1s)
Finding VCD file...
./dump.vcd
[2026-05-15 05:05:24 UTC] Opening EPWave...
Done
```

Task 2 (ram design)

EPWave

EDA playground

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Languages & Libraries

Testbench + Design

SystemVerilog/Verilog

UVM / OVM

None

Other Libraries

None OVL SVUnit

Enable TL-Verilog

Enable Easier UVM

Enable VUnit

Tools & Simulators

Icarus Verilog 12.0

Compile Options

-Wall -g2012

Run Options

Run Options

Use run.bash shell script

Open EPWave after run

Show output text file

Show output SVG file

Download files after run

Examples

testbench.sv

```
1 module testbench;
2
3 reg clk;
4
5 pipeline_processor uut(
6     .clk(clk)
7 );
8
9 always #5 clk = ~clk;
10
11 initial begin
12     $dumpfile("dump.vcd");
13     $dumpvars(0, testbench);
14
15     clk = 0;
16
17     #50;
18     $finish;
19 end
20 endmodule
```

design.sv

```
1 reg [7:0] EX_stage, WB_stage;
2 always @(posedge clk)
3 begin
4     IF_stage <= opcode;
5     ID_stage <= IF_stage;
6     case(ID_stage)
7     2'b00: EX_stage <= A + B;
8     2'b01: EX_stage <= A - B;
9     2'b10: EX_stage <= 8'd20;
10    default: EX_stage <= 0;
11    endcase
12    WB_stage <= EX_stage;
13    result <= WB_stage;
14 end
15 initial
16 begin
17     opcode = 2'b00;
18     A = 8'd15;
19     B = 8'd5;
20 end
21 endmodule
```

Log Share

[2026-05-15 06:11:12 UTC] iverilog -wall -g2012 design.sv testbench.sv && unbuffer vvp a.out

VCD info: dumpfile dump.vcd opened for output.

testbench.sv:19: \$finish called at 50 (1s).

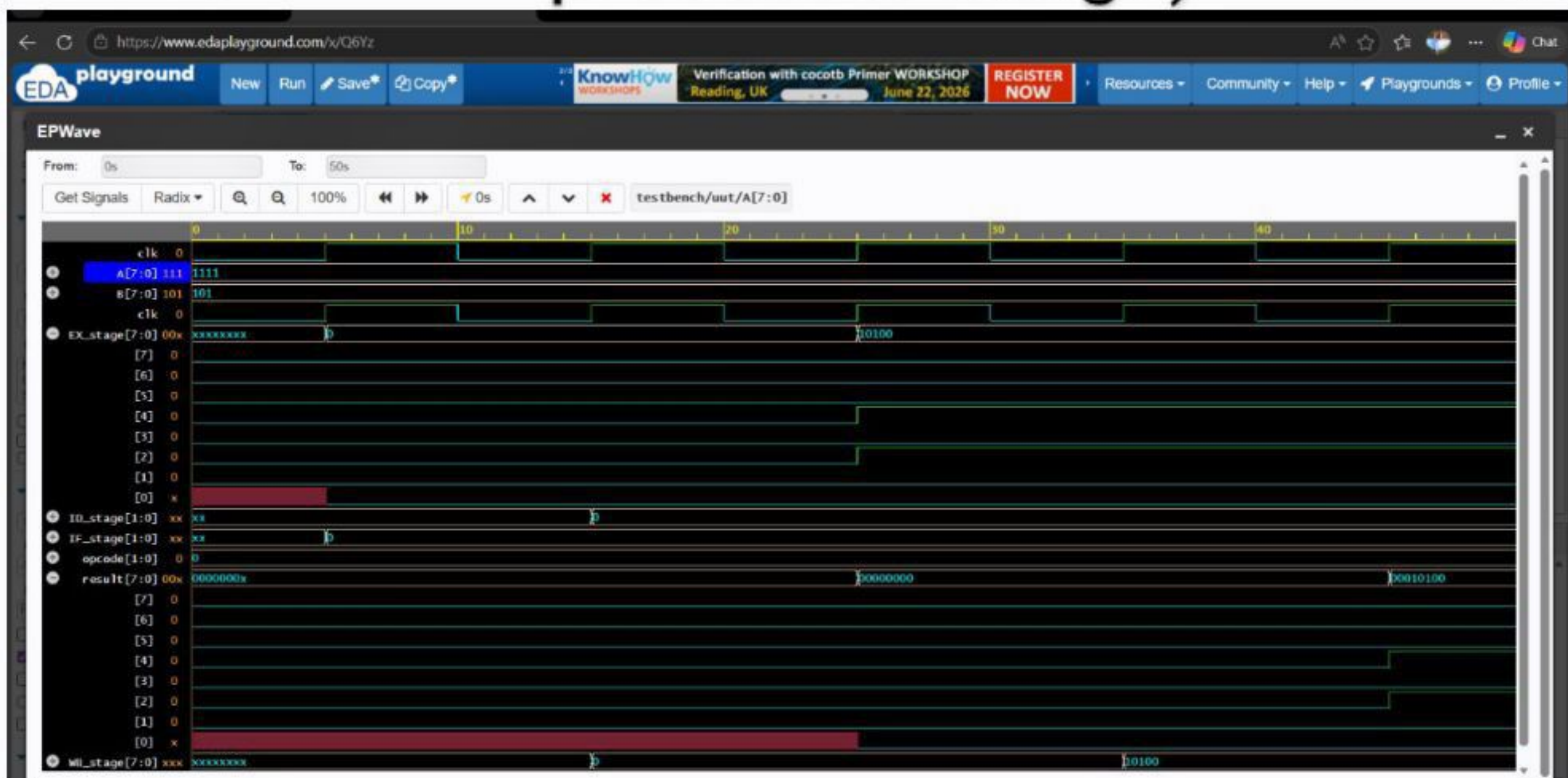
Finding VCD file...

./dump.vcd

[2026-05-15 06:11:13 UTC] Opening EPWave...

Done

Task 3(pipeline processor design)




```

% Normalize cutoff frequency
Wn = fc/(fs/2);
% FIR filter design (Hamming window)
b = fir1(N, Wn, 'low', hamming(N+1));
% Frequency response
figure;
freqz(b,1,1024,fs);
title('FIR Low Pass Filter Frequency Response');
% Input signal (signal + noise)
t = 0:1/fs:1;
x = sin(2*pi*50*t) + sin(2*pi*300*t);
% Apply filter
y = filter(b,1,x);
% Time domain plot
figure;
plot(t,x,'b'); hold on;
plot(t,y,'r','LineWidth',1.5);
legend('Input Signal','Filtered Signal');
title('Digital FIR Filter Output');
xlabel('Time');
ylabel('Amplitude');

```

 N	20	1×1	⋮
 t	1×1001 double	1×1001	⋮
 Wn	0.2000	1×1	⋮
 x	1×1001 double	1×1001	⋮
 y	1×1001 double	1×1001	⋮

Task 4(digital filter design)

